

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>

To find the de Broglie wavelength , we use the formula $\lambda = h / (\text{sqrt}(2 m E))$, where h is Planck's constant , m is the mass of the electron , and E is the kinetic energy . We need to convert 1 eV to joules for the calculation . $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$. The mass of an electron is $9.109 \times 10^{-31} \text{ kg}$. Plugging in the values :

$\lambda = (6.626 \times 10^{-34} \text{ J*s} / (\text{sqrt}(2*9.109 \times 10^{-31} \text{ kg} *1.602 \times 10^{-19} \text{ J}))) = 2.427 \times 10^{-10} \text{ m} \hat{=} 2.48 \times 10^{-8} \text{ cm}$

The closest answer to this calculation is $2.48 \times 10^{-8} \text{ cm}$.

</think>

<answer> $2.48 \times 10^{-8} \text{ cm}$ </answer>